

TESTING PLAN

Prison Management System

1. Purpose of Testing

The purpose of testing is to ensure that the **Prison Management System** works correctly, securely, and efficiently according to the system requirements. Testing helps identify errors, validate functionalities, and confirm that the system solves real-world prison management problems.

2. Testing Scope

The testing covers the following system modules:

- User Authentication (Login & Roles)
- Prisoner Management
- Cell and Block Assignment
- Visitor Management
- Medical Records
- Incident Management
- Report Generation

3. Testing Types Used

3.1 Unit Testing

Each module was tested individually to ensure correct functionality.

Examples:

- Login module
- Prisoner registration module
- Visitor registration module

✓ Ensures each function works independently.

3.2 Integration Testing

Tests interaction between system modules and the database.

Examples:

- Prisoner registration + database storage
- Visitor approval + status update
- Medical record update + prisoner record

✓ Ensures modules work together correctly.

3.3 Validation Testing

Ensures correct data is entered into the system.

Examples:

- Empty input fields
- Invalid prisoner ID
- Invalid login credentials

✓ Prevents incorrect or incomplete data.

3.4 Functional Testing

Tests whether system features meet functional requirements.

Examples:

- Register prisoner
- Assign cell
- Record medical checkup
- Generate reports

✓ Confirms system performs required tasks.

3.5 User Acceptance Testing (UAT)

End-user testing to confirm the system is usable in a real prison environment.

Actors involved:

- Admin
- Prison Officer

- Medical Staff
- Visitor Officer

✓ Confirms system meets user expectations.

4. Test Cases

Test Case 1: Login Authentication

Item	Description
Test Case ID	TC-01
Module	Login
Input	Wrong username/password
Expected Result	Error message displayed
Actual Result	Error message displayed
Status	Passed

Test Case 2: Register Prisoner

Item	Description
Test Case ID	TC-02
Module	Prisoner Management
Input	Valid prisoner details
Expected Result	Prisoner registered successfully
Actual Result	Prisoner registered
Status	Passed

Test Case 3: Assign Prisoner to Cell

Item	Description
Test Case ID	TC-03
Module	Cell Assignment
Input	Available cell
Expected Result	Cell assigned and capacity updated
Actual Result	Cell assigned
Status	Passed

Test Case 4: Visitor Approval

Item	Description
Test Case ID	TC-04
Module	Visitor Management
Input	Approve visitor request
Expected Result	Visitor status updated
Actual Result	Status updated
Status	Passed

Test Case 5: Medical Record Entry

Item	Description
Test Case ID	TC-05
Module	Medical Records
Input	Valid diagnosis

Item	Description
	and treatment
Expected Result	Medical record saved
Actual Result	Record saved
Status	Passed

5. Testing Environment

- Operating System: Windows
- Server: XAMPP
- Database: MySQL
- Browser: Chrome / Firefox

6. Test Results Summary

All test cases were executed successfully.

No critical errors were found.

The system meets all functional requirements and performs as expected.

7. Conclusion

The Prison Management System was thoroughly tested using different testing techniques. The testing process confirms that the system is **reliable, secure, and ready for real-world use in prison administration.**